

SARDAR MD. SAFFAT ZABIN

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EDUCATION

Bangladesh University of Engineering & Technology

B.Sc. in Computer Science and Engineering; **CGPA: 3.59/4.00**

Dhaka, Bangladesh

Feb 2020 – Mar 2025

Notre Dame College

Higher Secondary Certificate (HSC); **GPA: 5.00/5.00**

Dhaka, Bangladesh

2017 – 2019

RESEARCH INTERESTS

Computer Architecture & Systems, Database, Machine Learning, Human Computer Interaction

RELEVANT COURSEWORK

Major coursework: Object Oriented Programming, Discrete Math, Digital Logic Design, Data Structures & Algorithms, Database, Theory of Computation, Compiler, Computer Architecture, Microprocessors, Data Communication, Software Engineering, Information System Design, Artificial Intelligence, Operating System, Computer Networks, Computer Security, Bioinformatics, Computer Graphics, Basic Graph Theory, Algorithm Engineering, Machine Learning

Minor coursework:

Electrical Engineering: Basic Electrical Engineering, Electronic Circuits, Electrical Drives and Instrumentation

Mathematics: Differential Calculus, Integral Calculus, Coordinate Geometry, ODE, PDE, Vector Calculus, Complex Variables, Statistics, Probability, Linear Algebra, Laplace Transformation, Fourier Analysis

Mechanical Engineering: Basic Mechanical Engineering, AutoCAD

Physics: Heat & Thermodynamics, Electricity & Magnetism, Waves & Oscillation, Quantum Mechanics

Chemistry: Chemistry basics, Computational Chemistry

Others: English, Economics, Accounting, Industrial & Production Engineering, Government Studies

WORK EXPERIENCE

OPlusDojo (Remote)

Software Engineer

Toronto, Canada

Sep 2025 – Feb 2026

- Built an asynchronous document ingestion and **RAG** pipeline using **FastAPI**, **Celery**, **Redis**, **S3**, and **PostgreSQL/pgvector** to process uploaded files, extract structured content and embeddings via **Gemini**, and enable semantic search with LLM-evaluated answer generation.
- Built and optimized backend APIs using **RDS PostgreSQL + pgvector** and **S3 presigned URLs**; reduced **p95 latency from 1851 ms → 30 ms (98.2%)** through endpoint restructuring; implemented JWT based authentication using **AWS Cognito**.
- Designed and maintained **development CI/CD pipelines** using GitHub Actions for Docker builds, image tagging, and automated deployments to AWS dev environments.
- Worked on deployment workflows for Dockerized **FastAPI** services in AWS environments; configured two **Application Load Balancers (ALB)** (one for API service, one for MCP service) routing behind **CloudFront** for production traffic handling.
- Built internal dashboard views and reusable components in **React + TypeScript** for platform workflows ([ralfh-dev.com](https://github.com/ralfh-dev)).

Apurba Technologies Ltd.

Full Stack Developer (L-1)

Dhaka, Bangladesh

Mar 2025 – Oct 2025

- Built a **JavaFX** desktop tool to convert legacy ASCII Bangla fonts to Unicode with concurrent processing (ExecutorService), progress tracking, and cancellation.
- Extended the tool into a full-stack web platform using **Spring Boot** (backend) and **Angular** (frontend); deployed via **Docker Compose** on a government server (ocr.bangla.gov.bd).
- Designed a configurable cron-based cleanup scheduler in **Spring Boot**, driven by YAML configuration to invalidate and delete temporary OCR files with environment-specific policies (e.g., hourly in dev, weekly at 3AM in prod), eliminating manual storage maintenance.

RESEARCH EXPERIENCE

Independent Research Collaboration

Researcher, NoSQL Database Data Structures

Dhaka, Bangladesh

January 2026 – Present

- Conducting independent research under [Dr. Tarikul Islam Papon](#) on a **physics-informed reinforcement learning compaction trigger** for LSM-tree key-value stores, combining an analytic LSM cost prior with a learned residual so the policy adapts to workload shifts online from a cold start, reducing read and write amplification while RocksDB retains file selection and correctness.

Independent Research Collaboration

Researcher, Computer Vision

Dhaka, Bangladesh

October 2025 – January 2026

- Conducting independent research under the supervision of [Dr. M. Saifur Rahman](#) on a **compute-efficient Bengali image captioning pipeline** combining a frozen MaxViT encoder with an mBART-50 Bengali-native decoder, proposing a tri-loss framework (*Patch-Alignment Loss*, *InfoNCE*, *Sinkhorn OT*) and building a Bengali-aligned corpus with 110k bilingual-prompted synthetic images.
- Achieved state-of-the-art results on low-resource benchmarks (**BLEU-4 (12.29)**, **METEOR (27.98)**, **BERTScore (71.20)**) while narrowing the real-synthetic domain gap by 41%, demonstrating strong cross-domain generalization and single-GPU training efficiency for Bengali vision-language alignment.

Bangladesh University of Engineering & Technology

Undergraduate Researcher, Human-Computer Interaction

Dhaka, Bangladesh

Sep 2024 – March 2025

- Investigating the impact of government-imposed Internet shutdowns on Bangladeshi citizens during the July protests, guided by [Dr. Alim Al Razi](#) and [Dr. Novia Nurain](#).
- Developed a **transcription pipeline** using OpenAI's Whisper model for Bangla-to-English speech translation and Pyannote's diarization model for speaker segmentation. Preprocessed audio files, optimized Whisper for high accuracy translations, and achieved efficient speaker diarization for 2-3 speakers in long-format interviews. Conducted Performance evaluation in real-world interview datasets to validate accuracy and robustness.
- Developed a pipeline using [RoBERTa based GoEmotion](#) model and [DistilRoberta](#) based model for sentiment classification from the transcribed English text. The RoBERTa model which is fine-tuned and trained using the [GoEmotions Dataset](#) can detect upto 27 sentiments and quantify those sentiments as well.

Bangladesh University of Engineering & Technology

Undergraduate Thesis Supervisee, Machine Learning for Systems

Dhaka, Bangladesh

Nov 2023 – March 2025

- Reproduced the perceptron-based prefetch filter of Bhatia et al. ([Perceptron-based Prefetch Filtering](#), ISCA 2019) through an independent C++ implementation layered over the Signature Path Prefetcher, and evaluated its prefetch coverage, accuracy, and IPC against the reported results using the [ChampSim](#) trace-based simulator on the SPEC CPU2017 suite.
- Proposed and implemented a feed-forward neural network filter, employing a softmax output layer with cross-entropy loss and online weight updates, as a substitute for the perceptron. By dispensing with the one-hot encoded weight tables required by the perceptron formulation, the proposed design **reduces model weight storage by 93% (22.13 kB to 1.45 kB) and total filter memory overhead by 47%**, while incurring less than 1% degradation in IPC relative to the perceptron baseline.

PUBLICATIONS

- First Author.** Beyond the Blackout: Qualitative Insights and Sentiment Analysis on the Impact, Adaptation, and Design Solutions During the Internet Shutdown in Bangladesh. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Acceptance rate 32.83%. [\[Paper\]](#)
- Second Author.** Align Where the Words Look: Cross-Attention-Guided Patch Alignment with Contrastive and Transport Regularization for Bengali Captioning. **Submitted to PRICAI 2026**. [\[Paper\]](#)

ACADEMIC & OTHER PROJECTS

Brain Tumor Classification using Vision Transformer | [GitHub](#)

- A Image Processing task where we leveraged the newly developed Vision Transformer architecture to evaluate and bring further improvements to the brain tumor classification tasks over the conventional CNN architectures. We chose the paper [A Deep Analysis of Brain Tumor Detection from MR Images Using Deep Learning Networks](#) to evaluate the results of this paper by implementing a Vision Transformer Architecture, namely [Cross-ViT](#), which is discussed in the paper titled [CrossViT: Cross-Attention Multi-Scale Vision Transformer for Image Classification](#).

Machine Learning Algorithms | [GitHub](#)

- In Machine Learning Sessional, Logistic Regression, Bagging Ensemble learning, Stacking Ensemble learning, Feed Forward Neural Network, EM Algorithm and Principal Component Analysis has been implemented from scratch
- **Language:** Python
- **Libraries:** Pandas, Numpy, Scikit-learn

BERT Sentiment Classification | [GitHub](#)

- A Natural Language Processing (NLP) Project where BERT model was used to classify sentiments from tweets published during the Covid-19 pandemic. The project was made using Pytorch and uses BERT Tokenizer, BERT Sequence Classifier, gensim, Numpy, Pandas, Scikit-learn among other libraries.

UrbanNest: Real Estate Platform | [Live](#) | [Backend](#) | [Frontend](#)

- Designed a layered **Spring Boot** backend with DTO separation, **MapStruct** mapping, **OpenAPI** documentation, and **Flyway**-managed schema evolution on **PostgreSQL (AWS RDS)**.
- Implemented dynamic search using **JPA Specifications** and optimized page queries with **@EntityGraph**, reducing N+1 queries from 81+ per page to 2-3 (**97.5% improvement**).
- Built secure authentication using **Firebase JWT + Spring Security**; protected sensitive data with **AES-256-GCM encryption** and **SHA-512 hashing**; implemented file uploads via **AWS S3 presigned URLs**.
- Integrated **Google Maps Platform** (Maps JS, Places API, Geocoding API) for interactive location selection, reverse geocoding, and nearby place search with client-side caching and radius overlays in the UI (**React + TypeScript**).
- Deployed containerized services using **Docker** via **GitHub Actions** → **ECR** → **EC2**, with **Nginx reverse proxy** and **Let's Encrypt SSL termination**; frontend delivered through **S3 + CloudFront (ACM)** CDN with monitoring via **Prometheus + Grafana**.

Computer Security | [GitHub](#)

- In this project, Symmetric Key cryptography - AES encryption system with 128, 192 and 256 byte key and Public Key cryptography - Diffie Hellman key exchange, Cross Site Scripting and Buffer Overflow are implemented.
- **Language:** Python

Operating System | [GitHub](#)

- This project is a step by step enhancement of a teaching operating system named xv6 where new system calls, Inter Process Communication, Memory management (copy on write, paging) are implemented into the source code of xv6.
- **Language:** Bash, C

Computer Networks | [GitHub](#)

- This coursework includes Java implementation of Socket Programming, Building topologies in NS3 and Implementation of the TCP-Reno congestion control algorithm in NS3, as well as Packet Tracer simulation
- **Language:** C++, Java

Computer Architecture Sessional | [GitHub](#)

- Designed and simulated digital circuits, a 32 bit floating point adder and subtractor, and a 4-bit processor for executing MIPS pipeline in logisim, using basic gates and other ICs

Micro-controller Sessional | [GitHub](#)

- Built a game *Maguire In The Middle* which is controlled by an ATmega32A and an Arduino UNO. The game screen is a 16x16 bicolorable LED Dot Matrix, created using 4 8x8 bicolorable LED Dot Matrix and 6 decoder IC's. The time and points are shown using a 16x2 LCD display module. 4 Joysticks are used to control 4 paddles around the screen, and a gyroscope (MPU 6050) is used to direct the movement of the Maguire.

Artificial Intelligence Sessional | [GitHub](#)

- In this sessional, we implemented various classical Artificial Intelligence algorithms, such as A* search, Adversarial Search using alpha-beta pruning, Supervised Learning using Decision Tree etc.
- **Language:** C++

C Compiler | [GitHub](#)

- An incremental development of a compiler for a subset of C Programming language
- **Language:** C

Litol Football Manager | [Github](#)

- Aimed at creating a seamless experience at managing a football team, along with player transfers. Inspired by Football Manager game series.
- **Tech Stack:** JavaFx, Java Networking, Java Threading

SKILLS

Programming Languages: Java, C/C++, Python, TypeScript, SQL
Web & Application Frameworks: Spring Boot, FastAPI, Celery, React, Node.js, Express.js, JavaFX
Databases: PostgreSQL (with pgvector), OracleDB, MySQL
Cloud & Infrastructure: AWS (Cognito, EC2, RDS, S3, CloudFront, ECR, ACM), Google Maps Platform (Maps JS, Places API, Geocoding API), Redis, Nginx, Firebase
DevOps: Docker, Docker Compose, GitHub Actions
AI/LLM: LangChain, LangGraph, Gemini API
Version Control: Git, GitHub
Operating Systems: Linux (Debian 13, Ubuntu 22.04), Windows
Tools & Misc.: OSSEC-HIDS, L^AT_EX (Beamer / Overleaf), Figma, xv6, ns-3, yacc/bison/flex, Arduino (ATmega32). Maven, Gradle

AWARDS & ACHIEVEMENTS

RISE Undergraduate Student Researcher Grant: Awarded to students with promising thesis projects for implementing their proposals by the Research & Innovation Center for Science & Engineering (RISE) institute of Bangladesh University of Engineering & Technology. (Nov 2023 – March 2025)
Poster acceptance at NSysS 2024: My poster on the research work regarding Human-Computer Interaction was accepted in the NSysS 2024 conference

LEADERSHIP ACTIVITIES

Publication Committee Chair <i>IEEE Computer Society BUET Student Branch Chapter</i>	<i>March 2024 – March 2025</i>
Assistant Treasurer <i>IEEE Computer Society BUET Student Branch Chapter</i>	<i>March 2023 – March 2024</i>
Programming Contest Coordinator <i>IEEE Computer Society BUET Student Branch Chapter</i>	<i>February 2022 – March 2023</i>

REFERENCES

Dr. Rifat Shahriyar Professor , Department of CSE Bangladesh University of Engineering & Technology (BUET) Email: rifat.shahriyar@gmail.com rifat@cse.buet.ac.bd Relation: Thesis Advisor, Research Advisor	Dr. Tarikul Islam Papon Assistant Professor , Department of CS University of Massachusetts, Boston Email: t.papon@umb.edu Relation: Research Advisor
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